Toward AUV-Assisted Optical-Acoustic Detection of Invasive Mystery Snails

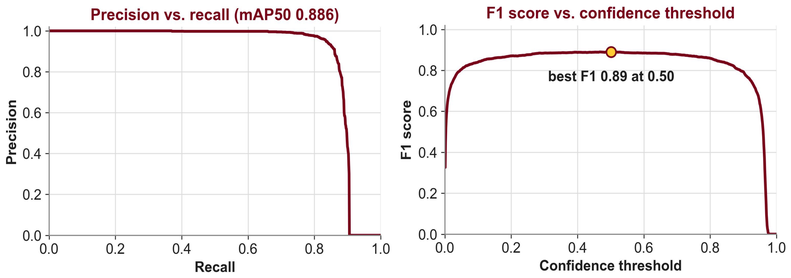
Amreen Hossain, Sander Versteele, Junaed Sattar · Minnesota Robotics Institute (MnRI), Department of Computer Science and Engineering, University of Minnesota Twin Cities

1 Detector on labeled photos

YOLO11s, pretrained on COCO, fine-tuned for 50 epochs on about 6,900 iNaturalist snail photos. Tested on 782 held-out photos with 1,324 snails.

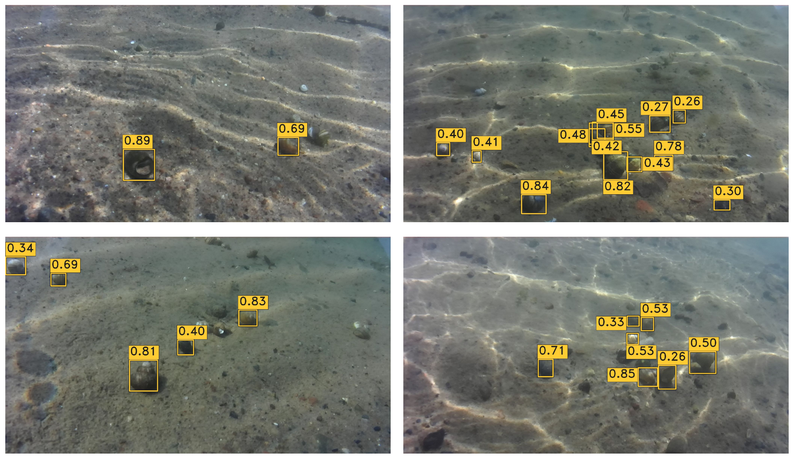
0.95	0.84	0.89	0.89	0.80
Precision	Recall	F1 score	mAP50	mAP50-95





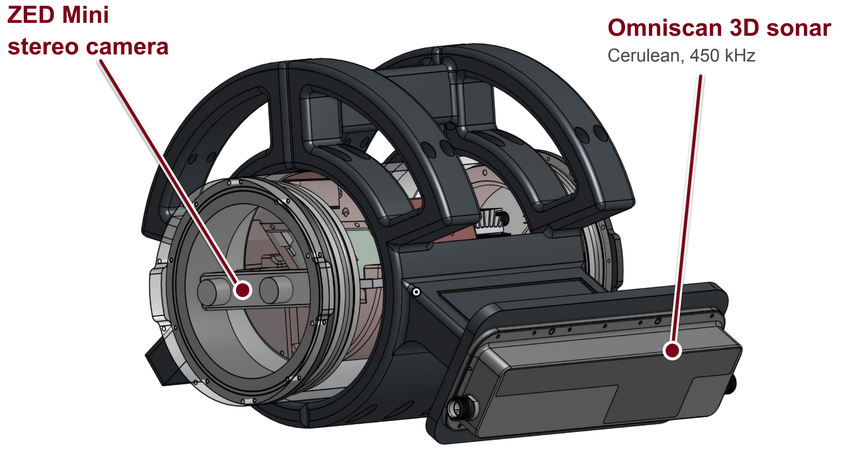
2 On our robot's lake footage

Run as-is on frames from our 2026 lake trials; it was never trained on robot footage. We are labeling these frames to measure its accuracy and to fine-tune it for our robot.



Yellow boxes: predicted snail detections. Numbers indicate detection confidence (0–1).

Can an underwater robot find invasive mystery snails with camera and sonar?



HydroEye, our lab's diver-held robot, with its stereo camera and 3D sonar.

Chinese mystery snails (*Cipangopaludina chinensis*) are invasive in Minnesota lakes, and surveys still rely on divers searching manually. Cameras see only a short distance in murky lake water, while sonar sees through it but gives coarse, noisy images. We are combining the two on our robot.



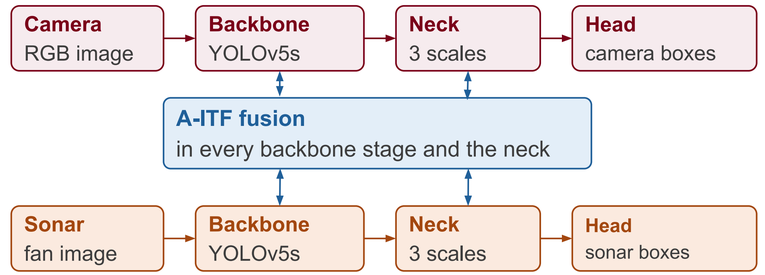
Square Lake, August 2026: HydroEye on the shore, the team deploying it, and a mystery snail.

Roadmap

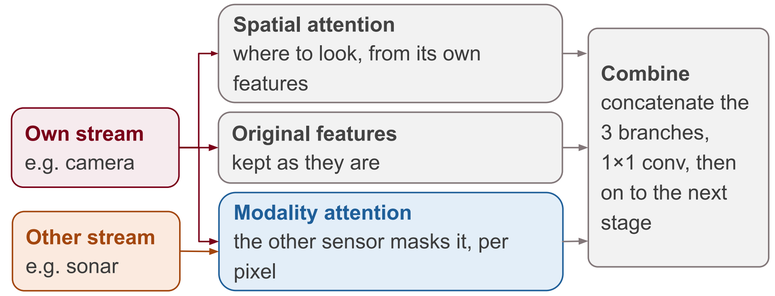
- **Ongoing:** Build a detection pipeline that fuses stereo depth with 3D sonar
- **Next:** Collect a paired sonar-camera dataset of mystery snails
- **Future:** Deploy the pipeline on an AUV to detect and localize snails in lakes

3 Sonar-camera fusion (AO-UOD)

We implement and apply the AO-UOD model [1].



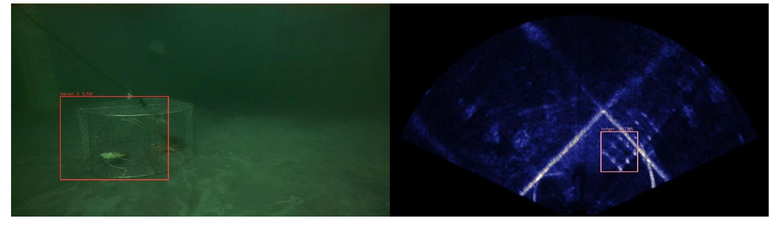
Inside A-ITF (attention-based interactive dual-branch fusion), for one stream



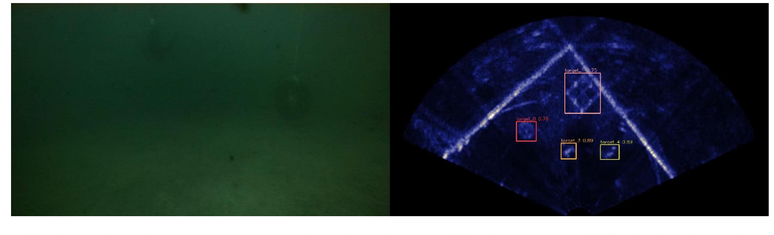
Trained and tested on UMOD [2]: 259 camera-sonar pairs from an ROV in very turbid water, 9 object classes.

mAP50 on the UMOD test set

0.41	0.66	0.54	8 ms
camera head	sonar head	both heads	per pair (GPU)



Both heads find the same trap: camera (left), sonar (right).



Murky water: the camera sees almost nothing, while the sonar still finds the targets.

References:
[1] Yu et al., "AO-UOD: A Novel Paradigm for Underwater Object Detection Using Acousto-Optic Fusion," IEEE Journal of Oceanic Engineering, 2025.
[2] UMOD dataset, github.com/UM-Research-Ng/UMOD-dataset.

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